# OS ORANGE REPORT

Q5.1:To implement pes checkout <branch>, the .pes/HEAD file must be updated to point to the new branch reference (e.g., ref: refs/heads/<branch>). The working directory must be updated by deleting files that are not in the new branch's commit tree, creating files that are new to the branch, and replacing modified files with their respective blobs from the new branch's tree. The index (.pes/index) must also be completely overwritten to match the new tree. This operation is complex because you must safely update the filesystem without accidentally deleting the user's uncommitted work, which requires carefully compariNg the current working directory, the index, and the target branch tree before making any destructive changes.

Q5.2:To detect a dirty working directory, you compare the current index against both the working directory and the target branch's tree. First, hash the files in the working directory and compare them to the hashes stored in the .pes/index. If they differ (or if a file is missing/untracked), the file has uncommitted changes. Second, compare the index entries against the tree objects of the target branch's commit. If an uncommitted file in the working directory differs from the blob hash that the target branch expects to place there, a conflict exists and the checkout must refuse to proceed to prevent overwriting the user's work.

Q5.3: When making commits in a "Detached HEAD" state, the commits are successfully created in the object store, and HEAD is updated to point to the newest commit hash. However, because no branch file (like refs/heads/main) is updated, the commits are not attached to any named branch. If you checkout another branch later, those commits become unreachable "orphans". A user can recover these commits by finding their hash (via terminal history or the reflog in actual Git) and creating a new branch that points to that specific hash (e.g., git branch my-recovered-commits <hash>).

Q6.1:The algorithm to find and delete unreachable objects is a Mark-and-Sweep garbage collection. First, you initialize an empty Set data structure (like a hash set) to track "reachable" hashes. You start at the "roots" by reading all branch files in .pes/refs/heads/ and the .pes/HEAD file. For each commit hash found, you traverse its history (parents) and mark them as reachable in the Set. Then, for each reachable commit, you traverse its tree object and all sub-trees, marking every referenced blob and tree hash as reachable. Finally, you

```
iterate through every actual file in .pes/objects/; if a file's hash is
not in the Set, it is unreachable and is deleted. For a repository with
100,000 commits and 50 branches, you would theoretically need to visit
at least 100,000 commits plus the trees and blobs they reference,
likely resulting in millions of object visits (though caching and
deduplication drastically reduce actual disk reads).

 Q6.2: It is dangerous because of race conditions between object
creation and reference updating. Suppose a user runs pes add file.txt,
creating a new blob in the object store. At this exact moment, a
concurrent GC runs. The GC scans the branch references and trees, but
since the new commit hasn't been created yet, the GC determines the new
blob is "unreachable" and deletes it. A split-second later, pes commit
finishes and tries to reference the blob, resulting in a corrupted
repository with a missing object. Git's real GC avoids this by using a
grace period (e.g., only deleting objects that are older than 2 weeks)
and relying on the modification timestamps of the objects, ensuring
that newly created but not-yet-referenced objects are safe
```

**{1A,1B}**

```
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc
12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e
5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
```

**{2A,2B}**

```
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

```
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ find .pes/objects -type f
.pes/objects/6a/b21510793a4c4b3dcb39401e79f7bb0dca3c74d1f42fc647ca296ded7b38df
.pes/objects/4d/29629e70a15b97ad3c33e17b1583b2db24e6baba6aeab2d93ba4dee092a0dc
.pes/objects/01/66224e3a7ff2a84e3bc93440e9deca5490699fa5516311b8fecc36102f3283
.pes/objects/4b/b39403abe83e593cdfb3de0c15e4d29622090094218e9858af6ce206755a93
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ xxd .pes/objects/6a/b21510793a4c4b3dcb39401e
79f7bb0dca3c74d1f42fc647ca296ded7b38df| head -20
00000000: 626c 6f62 2032 3500 696e 7420 6d61 696e  blob 25.int main
00000010: 2829 207b 2072 6574 7572 6e20 303b 207d  () { return 0; }
00000020: 0a                                       .
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ xxd .pes/objects/4d/29629e70a15b97ad3c33e17b
1583b2db24e6baba6aeab2d93ba4dee092a0dc| head -20
00000000: 636f 6d6d 6974 2032 3134 0074 7265 6520  commit 214.tree
00000010: 3462 6233 3934 3033 6162 6538 3365 3539  4bb39403abe83e59
00000020: 3363 6466 6233 6465 3063 3135 6534 6432  3cdfb3de0c15e4d2
00000030: 3936 3232 3039 3030 3934 3231 3865 3938  9622090094218e98
00000040: 3538 6166 3663 6532 3036 3735 3561 3933  58af6ce206755a93
00000050: 0a61 7574 686f 7220 4172 7961 6e20 5265  .author Aryan Re
00000060: 7661 6e6b 6172 203c 5045 5331 5547 3234  vankar <PES1UG24
00000070: 4353 3038 343e 2031 3737 3637 3034 3133  CS084> 177670413
00000080: 370a 636f 6d6d 6974 7465 7220 4172 7961  7.committer Arya
00000090: 6e20 5265 7661 6e6b 6172 203c 5045 5331  n Revankar <PES1
000000a0: 5547 3234 4353 3038 343e 2031 3737 3637  UG24CS084> 17767
000000b0: 3034 3133 370a 0a47 656e 6572 6174 696e  04137..Generatin
000000c0: 6720 6120 7472 6565 206f 626a 6563 7420  g a tree object
000000d0: 666f 7220 5363 7265 656e 7368 6f74 2032  for Screenshot 2
000000e0: 42                                       B
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$
```

**{3A,3B}**

```
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ make pes
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ echo "hello" > file1.txt
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ echo "world" > file2.txt
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ ./pes add file1.txt file2.txt
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_tree
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  index.c
  untracked:  tree.h
  untracked:  index.h
  untracked:  commit.h
  untracked:  README.md
  untracked:  .gitignore
  untracked:  commit.c
  untracked:  pes.h
  untracked:  object.c
  untracked:  pes.c
  untracked:  tree.c
  untracked:  Makefile
  untracked:  test_sequence.sh

aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776701051 6
file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776701057 6
file2.txt
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$
```

**{4A,4B}**

```
./pes init
Initialized empty PES repository in .pes/
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"
Committed: 1d63464da03d... Initial commit
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"
Committed: 34cecf03bd1c... Add world
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"
Committed: 7cc85fc1dfc7... Add farewell
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ ./pes log
commit 7cc85fc1dfc707d7bf51fc5451ef02e165957e008f36d2c43d9cf1573733e7c8
Author: Aryan Revankar <PES1UG24CS084>
Date:   1776701841

    Add farewell_cert_id!

commit 34cecf03bd1c16052328dbf69275912a355799b7a4f7f9db59438a0c8504d97d
Author: Aryan Revankar <PES1UG24CS084>
Date:   1776701833

    Add world

commit 1d63464da03d45a8bfd7bec566a74be8bccb0a4de1c816850fff360d69d9ad54
Author: Aryan Revankar <PES1UG24CS084>
Date:   1776701811

    Initial committial commitES1A

aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/1d/63464da03d45a8bfd7bec566a74be8bccb0a4de1c816850fff360d69d9ad54
.pes/objects/34/cecf03bd1c16052328dbf69275912a355799b7a4f7f9db59438a0c8504d97d
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/7c/c85fc1dfc707d7bf51fc5451ef02e165957e008f36d2c43d9cf1573733e7c8
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ cat .pes/HEAD
cat .pes/refs/heads/main
ref: refs/heads/main
7cc85fc1dfc707d7bf51fc5451ef02e165957e008f36d2c43d9cf1573733e7c8
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ git add commit.c
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ git commit -m "Finish Phase 4 implementation
```

**INTEGRATION TEST**

```
aryan_r@macbook: ~/PES1UG
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: bca2596a7fe1... Initial commit

Log after first commit:
commit bca2596a7fe15b2ac7aeaf927679f801c54f7ca765e793c36f8b1d6c54fb5cfb
Author: Aryan Revankar <PES1UG24CS084>
Date:   1776702033

    Initial commit


--- Second Commit ---
Committed: 46c61a2b4687... Update file.txt

--- Third Commit ---
Committed: 72334d961c98... Add farewell

--- Full History ---
commit 72334d961c98e5f90ce4c2fd802051eac5436d7e0835b6ef030f9d79b7ee4221
Author: Aryan Revankar <PES1UG24CS084>
Date:   1776702033

    Add farewell_cert_id!

commit 46c61a2b468763ac9d600bf8970ad5e6244f0e51cb962f9291e5ced660e5e0c0
Author: Aryan Revankar <PES1UG24CS084>
Date:   1776702033

    Update file.txtrt_id!
```

```
aryan_r@macbook: ~/PES1UG
Date:   1776702033

    Initial commit


--- Second Commit ---
Committed: 46c61a2b4687... Update file.txt

--- Third Commit ---
Committed: 72334d961c98... Add farewell

--- Full History ---
commit 72334d961c98e5f90ce4c2fd802051eac5436d7e0835b6ef030f9d79b7ee4221
Author: Aryan Revankar <PES1UG24CS084>
Date:   1776702033

    Add farewell_cert_id!

commit 46c61a2b468763ac9d600bf8970ad5e6244f0e51cb962f9291e5ced660e5e0c0
Author: Aryan Revankar <PES1UG24CS084>
Date:   1776702033

    Update file.txtrt_id!

commit bca2596a7fe15b2ac7aeaf927679f801c54f7ca765e793c36f8b1d6c54fb5cfb
Author: Aryan Revankar <PES1UG24CS084>
Date:   1776702033

    Initial committial commitES1A


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
72334d961c98e5f90ce4c2fd802051eac5436d7e0835b6ef030f9d79b7ee4221

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/46/c61a2b468763ac9d600bf8970ad5e6244f0e51cb962f9291e5ced660e5e0c0
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/72/334d961c98e5f90ce4c2fd802051eac5436d7e0835b6ef030f9d79b7ee4221
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa19744d901ce282f30e08de0ccd92
.pes/objects/bc/a2596a7fe15b2ac7aeaf927679f801c54f7ca765e793c36f8b1d6c54fb5cfb
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
aryan_r@macbook:~/PES1UG24CS084-pes-vcs$
```